

1. There are three charges, q , q and $-q$, which are all placed on x -axis. The coordinates for them are $x = a$, $x = 0$, and $x = -a$, respectively. For the far zone, $|x| \gg a$, and to the order of magnitude of dipole or a/x , what is the electric potential ϕ and field E on the point x ?

2. A long cylinder of radius R carries a charge density, $\rho = k r$, where k is a constant and r is the distance from the axis.
- (1) Find the electric field E inside and outside the cylinder.
 - (2) Now, let the cylinder be shielded coaxially with a metal shell of radius R' find the energy U of the electric field per unit length.

3. A parallel plate capacitor has area A and separation d . The upper half of the capacitor is filled with an insulator of relative dielectric constant ϵ_1 , and the lower half with another insulator of relative dielectric constant ϵ_2 . (1) Find the capacitance C of the capacitor. (2) Find the areal charge density σ of the interface between the two insulators if the capacitor is charged to a voltage V .

4. Find the self-inductance L of a toroidal coil with rectangular cross section (inner radius a , outer radius b , height h), which carries a total of N turns.



5. A soap bubble of radius r is in equilibrium with the outer air of pressure p . Now, it is slowly given a charge q , the radius of the bubble will increase gradually to R , due to the mutual repulsion of the surface charge. Regard the air as an ideal gas, and neglect the effect of the tension of the surface of the bubble.

(1) Find the pressure p_{field} of the electric field upon the bubble, the dielectric constant of the outer air being ϵ .

(2) Show that $q^2 = 32 \pi^2 \epsilon p R (R^3 - r^3)$.

6. There are two electromagnetic travelling waves overlapped in free space, their electric fields are $\mathbf{E}_1 = C \cos(\mathbf{k} \cdot \mathbf{r} - \omega t) \mathbf{e}_x$ and $\mathbf{E}_2 = D \cos(\mathbf{k} \cdot \mathbf{r} - \omega t) \mathbf{e}_x + D \cos(\mathbf{k} \cdot \mathbf{r} - \omega t) \mathbf{e}_y$, where $C > 0$ and $D > 0$ are constant, and ω and $\mathbf{k} = k \mathbf{e}_z$ are the frequency and wave vector ($\omega = c k$ where c is the speed of light), respectively. Here, x , y and z are Cartesian coordinates, and $\mathbf{e}_x$, $\mathbf{e}_y$ and $\mathbf{e}_z$ the corresponding unit vectors, t denotes the time, and $\mathbf{r} = x \mathbf{e}_x + y \mathbf{e}_y + z \mathbf{e}_z$.

(1) Find the total magnetic induction $\mathbf{B}$.

(2) Find the energy density u of the electromagnetic field and its time average $\langle u \rangle$ over one period.

(3) Find the Poynting vector $\mathbf{S}$ and its time average $\langle \mathbf{S} \rangle$ over one period.